

Ex 1.3

(13)

(Pg 18)

Q#1 Solve the following Simultaneous linear equations with Complex Coefficients for w and z .

$$\Rightarrow w(10+2i) = 28-14i$$

$$\Rightarrow w(5+i) = 14-7i$$

$$\Rightarrow w = \frac{14-7i}{5+i}$$

$$\Rightarrow w = \frac{14-7i}{5+i} \times \frac{5-i}{5-i}$$

$$\Rightarrow w = \frac{63-49i}{25+1}$$

$$\textcircled{1} 3z + (2+i)w = 11-i$$

$$(2-i)z - w = -1+i$$

$$\Sigma q \textcircled{1} \times (2-i) \quad 3(2-i)z + (2+i)(2-i)w = (11-i)(2-i)$$

$$\Sigma q \textcircled{2} \times 3 \quad 3(2-i)z - 3w = 3(-1+i)$$

$$w = \frac{63-49i}{26}$$

$$\Rightarrow (2+i)(2-i)w + 3w = (11-i)(2-i) - 3(-1+i)$$

$$\Rightarrow (2^2 - i^2)w + 3w = 22 - 11i - 2i + i^2 + 3 - 3i$$

$$\Rightarrow (4+1)w + 3w = 25 - 13i - 1 - 3i$$

$$\Rightarrow 8w = -16i + 24$$

$$\Rightarrow 8w = 8(-2i + 3)$$

$$\Rightarrow \boxed{w = 3-2i}$$

$$\text{Put in } 3z + (2+i)w = 11-i$$

$$\Rightarrow 3z + (2+i)(3-2i) = 11-i$$

$$\Rightarrow 3z + 6 - 4i + 3i - 2i^2 = 11-i$$

$$\Rightarrow 3z + 8 - i = 11-i$$

$$\Rightarrow 3z + 8 - i = 11-i$$

$$\Rightarrow 3z = 11-8$$

$$\Rightarrow 3z = 3$$

$$\boxed{z = 1}$$

$$2z + (3+i)w = 9-i$$

$$\Rightarrow 2z + (3+i)\left(\frac{63}{26} - \frac{49i}{26}\right) = 9-i$$

$$\Rightarrow 2z + \frac{1}{26}(238-84i) = 9-i$$

$$\Rightarrow 2z + \frac{119}{13} - \frac{42}{13}i = 9-i$$

$$\Rightarrow 2z = 9-i - \frac{119}{13} + \frac{42}{13}i$$

$$\Rightarrow 2z = \frac{117-13i-119+42i}{13}$$

$$\Rightarrow 2z = \frac{-2+29i}{13}$$

$$\boxed{z = \frac{-2}{26} + \frac{29i}{26}}$$

$$\text{Q#1 (ii)} 2z + (3+i)w = 9-i$$

$$(3-i)z - iw = -1+i$$

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$$\Sigma q \textcircled{1} \times (3-i) \Rightarrow 2(3-i)z + (3-i)(3+i)w = (3-i)(9-i)$$

$$\Sigma q \textcircled{2} \times 2 \Rightarrow 2(3-i)z - 2iw = 2(-1+i)$$

$$\Rightarrow (3-i)(3+i)w + 2iw = (3-i)(9-i) - 2(-1+i)$$

$$\Rightarrow (9-i^2)w + 2iw = 27-3i-9i+i^2+2-2i$$

$$\Rightarrow 10w + 2iw = 28-14i$$

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Q#2 Write the following Polynomials as the Product of linear factors.

(i) $z^2 + 8ib^2$

$$= z^2 + (9b)^2$$

$$= z^2 - (-i)(9b)^2$$

$$= z^2 - i^2(9b)^2$$

$$= z^2 - (9bi)^2$$

$$= (z - 9bi)(z + 9bi)$$

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(ii) $4z^2 + 9b^2$

$$= 4z^2 - (-1)9b^2$$

$$= 4z^2 - i^2 9b^2$$

$$= 4z^2 - (3bi)^2$$

$$= (2z)^2 - (3bi)^2$$

$$= (2z - 3bi)(2z + 3bi)$$

(ii) $z^3 + 3z^2 + z - 5$

Let $P(z) = z^3 + 3z^2 + z - 5$

is a Polynomial.

The factor of 5 are

$$\pm 1, \pm 5$$

$$P(1) = 1^3 + 3(1)^2 + (1) - 5 = 5 - 5 = 0$$

So $z - 1$ is a factor

$$\begin{array}{r|rrrr} 1 & 1 & 3 & 1 & -5 \\ & & 1 & 4 & 5 \\ \hline & 1 & 4 & 5 & 0 \end{array}$$

$$= (z - 1)(z^2 + 4z + 5)$$

$$\Rightarrow z^2 - 4z + 5 = 0$$

$$\Rightarrow z = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(5)}}{2(1)}$$

$$\Rightarrow z = \frac{-4 \pm \sqrt{16 - 20}}{2}$$

$$\Rightarrow z = \frac{-4 \pm \sqrt{-4}}{2}$$

$$\Rightarrow z = \frac{-4 \pm 2i}{2} = -2 \pm i$$

(iv) $z^3 + 3z^2 + 4z + 2$

Let $P(z) = z^3 + 3z^2 + 4z + 2$

is a Polynomial

The factor of 2 are

$$\pm 1, \pm 2$$

$$P(-1) = (-1)^3 + 3(-1)^2 + 4(-1) + 2$$

$$P(-1) = -1 + 3 - 4 + 2 = 0$$

So $z + 1$ is a factor

$$\begin{array}{r|rrrr} -1 & 1 & 3 & 4 & 2 \\ & & -1 & -2 & -2 \\ \hline & 1 & 2 & 2 & 0 \end{array}$$

$$(z + 1)(z^2 + 2z + 2)$$

$$\text{Now } z^2 + 2z + 2 = 0$$

$$\Rightarrow z = \frac{-2 \pm \sqrt{(2)^2 - 4(1)(2)}}{2(1)}$$

$$\Rightarrow z = \frac{-2 \pm \sqrt{4 - 8}}{2}$$

$$\Rightarrow z = \frac{-2 \pm \sqrt{-4}}{2}$$

$$\Rightarrow z = \frac{-2 \pm 2i}{2} = -1 \pm i$$

$$\Rightarrow (z + 1 - i)(z + 1 + i) = 0$$

$$= (z + 1)(z + 1 - i)(z + 1 + i) = 0$$

$$\textcircled{\text{iii}} \quad 34z^2 - 6z = -1$$

÷ by 34

$$\Rightarrow \frac{34}{34} z^2 - \frac{6}{34} z = -\frac{1}{34}$$

$$\Rightarrow z^2 - \frac{3}{17} z = -\frac{1}{34}$$

Adding $\left(\frac{a}{2p}\right)^2$ on Both side

$$\left(\frac{a}{2p}\right)^2 = \left(\frac{-\frac{3}{17}}{2(1)}\right)^2 = \left(-\frac{3}{17} \cdot \frac{1}{2}\right)^2 = \left(\frac{3}{34}\right)^2$$

$$\left(\frac{3}{34}\right)^2$$

$$\Rightarrow z^2 - \frac{3}{17} z + \left(\frac{3}{34}\right)^2 = -\frac{1}{34} + \left(\frac{3}{34}\right)^2$$

$$\Rightarrow \left(z - \frac{3}{34}\right)^2 = \frac{-34 + 9}{(34)^2} = \frac{-25}{(34)^2}$$

$$\Rightarrow \left(z - \frac{3}{34}\right)^2 = \frac{-25}{(34)^2}$$

$$\Rightarrow z = \pm \frac{5}{34} i + \frac{3}{34}$$

$$\Rightarrow \boxed{z = \frac{3}{34} \pm \frac{5}{34} i}$$

$$\textcircled{\text{iv}} \quad z^2 - 6z + 34 = 0$$

$$z^2 - 6z = -34$$

$$\left(\frac{a}{2p}\right)^2 = \left(\frac{-6}{2}\right)^2 = 9$$

$$\Rightarrow z^2 - 6z + 9 = -34 + 9$$

$$\Rightarrow (z - 3)^2 = -25$$

$$\Rightarrow z - 3 = \sqrt{-25}$$

$$\Rightarrow z - 3 = \pm 5i$$

$$\Rightarrow \boxed{z = 3 \pm 5i}$$

$$\textcircled{\text{v}} \quad z^2 - 6z = -13$$

$$\left(\frac{a}{2p}\right)^2 = \left(\frac{-6}{2}\right)^2 = 9$$

$$\Rightarrow z^2 - 6z + 9 = -13 + 9$$

$$\Rightarrow (z + 3)^2 = -4$$

$$\Rightarrow z - 3 = \sqrt{4}$$

$$\Rightarrow z - 3 = \pm 2i$$

$$\boxed{z = 3 \pm 2i}$$

$$\textcircled{\text{vi}} \quad z^2 + 64 = 0$$

$$z^2 = -64$$

$$\Rightarrow (z + 0)^2 = -64$$

$$\Rightarrow z + 0 = \sqrt{-64}$$

$$\Rightarrow z + 0 = \pm 8i$$

$$\Rightarrow \boxed{z = \pm 8i}$$

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Vii) $z^3 - 3z^2 + 4z - 2$ (Pg 16)

Let $P(z) = z^3 - 3z^2 + 4z - 2$
is a polynomial.

The factor of 2 is
 $\pm 1, \pm 2$

$$P(1) = 1^3 - 3(1)^2 + 4(1) - 2 = 0$$

So $z - 1$ is a factor $\Rightarrow z^2 - 4z + 2$

$$\begin{array}{r|rrrr} 1 & 1 & -3 & 4 & -2 \\ & & 1 & -2 & 2 \\ \hline & 1 & -2 & 2 & 0 \end{array}$$

$$\text{So } z^3 - 3z^2 + 4z - 2 = (z - 1)(z^2 - 2z + 2)$$

Now we can easily
find the factor of
 $z^2 - 2z + 2$

$$\text{let } z^2 - 2z + 2 = 0$$

$$\Rightarrow z = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(2)}}{2(1)}$$

$$\Rightarrow z = \frac{2 \pm \sqrt{4 - 8}}{2}$$

$$\Rightarrow z = \frac{2 \pm \sqrt{-4}}{2}$$

$$\Rightarrow z = \frac{2 \pm 2i}{2} = 1 \pm i$$

$$(z - 1)(z - 1 - i)(z - 1 + i) = 0$$

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Q#3 Solve the following
quadratic equations by
Completing The Squares.
where z is a Complex
number.

i) $z^2 - 4z + 5 = 0$

$$\Rightarrow z^2 - 4z = -5$$

Adding $\left(\frac{a}{2p}\right)^2$ on both

$$\text{side } \left(\frac{a}{2p}\right)^2 = \left(\frac{-4}{2}\right)^2 = 4$$

$$\Rightarrow z^2 - 4z + 4 = -5 + 4$$

$$\Rightarrow (z - 2)^2 = -1$$

$$\Rightarrow z - 2 = \sqrt{-1}$$

$$\Rightarrow z - 2 = \pm i$$

$$\Rightarrow \boxed{z = 2 \pm i}$$

ii) $z^2 + 12z + 52 = 0$

$$\Rightarrow z^2 + 12z = -52$$

$\Rightarrow$ Adding $\left(\frac{a}{2p}\right)^2 = \left(\frac{12}{2}\right)^2 = 36$
on Both side

$$\Rightarrow z^2 + 12z + 36 = -52 + 36$$

$$\Rightarrow (z + 6)^2 = -16$$

$$\Rightarrow z + 6 = \sqrt{-16}$$

$$\Rightarrow z + 6 = \pm 4i$$

$$\Rightarrow \boxed{z = -6 \pm 4i}$$

⑤ $z^3 - 7z^2 + 19z - 13$

let $P(z) = z^3 - 7z^2 + 19z - 13$
is a Polynomial

The factor of 13 are $\pm 1, \pm 13$

$P(1) = (1)^3 - 7(1)^2 + 19(1) - 13$

$P(1) = 1 - 7 + 19 - 13 = 0$

So $z-1$ is a factor

$$\begin{array}{r|rrrr} 1 & 1 & -7 & 19 & -13 \\ & & 1 & -6 & 13 \\ \hline & 1 & -6 & 13 & 0 \end{array}$$

$(z-1)(z^2 - 6z + 13)$

let $z^2 - 6z + 13 = 0$

$z = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(13)}}{2(1)}$

$z = \frac{6 \pm \sqrt{36 - 52}}{2}$

$z = \frac{6 \pm \sqrt{-16}}{2}$

$z = \frac{6 \pm 4i}{2}$

$z = 3 \pm 2i$

$(z-1)(z-3-2i)(z-3+2i) = 0$

⑥ $z^3 + 3z^2 + 19z + 17$
let $P(z) = z^3 + 3z^2 + 19z + 17$
is a Polynomial
The factor of 17 are $\pm 1, \pm 17$

For $z = +1$

$P(1) = 1^3 + 3(1)^2 + 19(1) + 17$

$\Rightarrow P(1) = 40 \neq 0$

For $z = -1$

$\Rightarrow P(-1) = (-1)^3 + 3(-1)^2 + 19(-1) + 17$

$\Rightarrow P(-1) = -1 + 3 - 19 + 17 = 0$

So $z+1$ is a factor

$$\begin{array}{r|rrrr} -1 & 1 & 3 & 19 & 17 \\ & & -1 & -2 & -17 \\ \hline & 1 & 2 & 17 & 0 \end{array}$$

Hence

$z^3 + 3z^2 + 19z + 17 = (z+1)(z^2 + 2z + 17)$

Now we can easily find the factor of $z^2 + 2z + 17$

$z^2 + 2z + 17 = 0$

$z = \frac{-2 \pm \sqrt{(2)^2 - 4(1)(17)}}{2(1)}$

$z = \frac{-2 \pm \sqrt{4 - 68}}{2} = \frac{-2 \pm 8i}{2}$

$z = -1 \pm 4i$

Thus $(z+1)(z+1-4i)(z+1+4i)$ are factors of

$z^3 + 3z^2 + 19z + 17$

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